

Effect of Binding Material on the Acoustic, Elastic and Strength Properties of Artificial Granular Rocks

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ABSTRACT: Artificial sandstone-like rocks fabricated in the laboratory offer great advantages in controlling their grain size distribution, grain morphology, porosity, and cement content. Synthetic granular rocks offer an efficient way to validate existing analytical models, calibrate numerical models, and perform sensitivity analyses on specific physical parameters. Synthetic sandstone rocks are manufactured by gluing grains together with a binding material. In this study, we investigated the effect of the three binding materials typically used in synthetic granular rock preparation on mechanical properties and acoustic wave velocities. We used ceramic spherical particles with sizes of $250\ \mu\text{m}$ - $425\ \mu\text{m}$ as grains bonded together by three binding materials, namely i) Epoxy resin, ii) Sodium Silicate with Kaolinite, and iii) Portland cement. While fabricating the core plugs, in each case we kept the relative volume proportion of the binding material with respect to the grains constant. These manufactured sandstone-like dry core plugs were then tested using Autolab-1500 triaxial testing equipment to measure the static mechanical properties and elastic wave velocities at a fixed confining pressure and temperature. Although grain size and distribution, confining pressure, etc., affect the mechanical behavior of rocks, our study focuses on the influence of the binding material alone. Despite the volume fraction of each binding material being the same, they influence the mechanical properties of the rock differently. The sample prepared with epoxy resin exhibited the highest values for Young's modulus, Poisson's ratio, and unconfined compressive strength (UCS) among the three samples. Our findings demonstrate that the selection of binding material plays an important role in controlling the mechanical properties of the rock. Two main reasons account for this behavior: i) the heterogeneous distribution of the binding material within the rock specimen and ii) the effect of the binding material itself on the contact behavior between the grains.

1 INTRODUCTION

Thermal-Hydro-Mechanical (THM) coupling processes are essential to various engineering and geoscience problems, such as geothermal energy extraction, geological storage of CO₂, and nuclear waste disposal. Understanding the interplay between these processes is crucial for predicting the behavior of the Earth's subsurface systems. However, the natural heterogeneity, inhomogeneity, and non-uniformity of geological materials make these processes inherently complex and difficult to model accurately (Tsang, 1991; Rutqvist and Tsang, 2005). To address this challenge, numerical modeling has become a popular tool for simulating THM processes. Nevertheless, robust calibration of numerical models remains a significant obstacle due to the natural variability of rocks (Yoon, 2007).

The influence of each physical constituent on THM pro-

cesses varies, and investigating these effects typically requires controlling all parameters except the one of interest. However, achieving this level of control over natural rocks is often challenging or even impossible due to the inherent variability of the material. Therefore, constructing synthetic rock analogs has emerged as a promising solution to this challenge. By creating synthetic rocks, researchers can better comprehend and evaluate the impact of various physical constituents on THM processes (Al-Homadhi, 2002).

The construction of synthetic rock analogs has several advantages over natural rocks. One of the primary benefits is the ability to precisely control the parameters of interest, such as grain size, shape, and cement quantity. Additionally, synthetic rocks can be designed to have simplified and idealized grain shapes, which are useful in calibrating numerical models, particularly discrete element methods

(Coetzee, 2020). Synthetic rocks can also be made with homogeneous compositions, making them more amenable to controlled laboratory experiments (Yin et al., 2017; Es-hiet and Sheng, 2017). Despite these benefits, constructing synthetic rocks is not without its challenges. Synthetic rocks must accurately reproduce the relevant physical and mechanical properties of natural rocks to ensure that their behavior mimics natural systems. Furthermore, the interdependence of various rock attributes, such as porosity and mineral composition, means that constructing synthetic rocks that accurately represent natural rocks can be difficult. Although there are challenges in accurately reproducing the complex physical and mechanical properties of natural rocks, synthetic rocks offer precise control over the parameters of interest and simplified grain shapes.

Previous researchers successfully constructed artificial sandstones using different types of materials and used them in different experimental analyses. These studies include mechanical behavior of fractured rocks e.g (Ding et al., 2020; Tillotson et al., 2014), petrophysical and geomechanical properties of granular rocks e.g (Shahsavari and Shakiba, 2022; Tien and Tsao, 2000; Förster et al., 1996), mechanical behavior of rock joints e.g (Hu et al., 2022; Tang and Wong, 2016) among others.

Artificial rocks were created with different kinds of materials popularly mixing sand grains with silica gel and kaolinite (Tillotson et al., 2012; Nakagawa and Myer, 2001; Al-Homadhi, 2002), sand grains mixed with cement (Santos et al., 2016, 2017), sand grains mixed with epoxy resin (Klimentos and Parker, 1988; Gomes et al., 2021). The choice of binding material used to create artificial rocks can have a significant impact on their physical and mechanical properties. For example, rocks created using sand grains mixed with silica gel and kaolinite exhibit different behavior from those created using sand grains mixed with cement or epoxy resin. Therefore, selecting an appropriate methodology for creating artificial rocks is dependent on the purpose of the study.

In this study, we aim to investigate the effect of different binding materials on the acoustic, mechanical, and strength properties of artificial granular rocks. The goal is to help researchers and engineers select an appropriate binding material depending on the purpose of their study. To achieve this goal, a series of experiments are conducted to compare the properties of artificial rocks created using different binding materials.

2 METHODOLOGY

In this research, firstly, we prepared synthetic rock specimens and then conducted mechanical experiments using the Autolab-1500 instrument.1.

2.1. Materials

The present study employed a ceramic proppant, composed of fused ceramic grains, adhered by a selected binding agent. The ceramic grains were procured from CARBO Ceramics Inc., USA, and marketed under the trade name Carbolite. The grains had a mesh size of 40/70, and 97% of their weight was made up of grains with diameters falling within the range of 250 μm to 425 μm . Furthermore, these grains were characterized by a spherical morphology, with a roundness and sphericity of 0.9 and a specific gravity of 2.7 g/cc. The manufacturer's product specifications are used to determine the size and shape of the grains' characteristic features.

The current investigation has selected three distinct binding materials based on their widespread usage in scientific research. Specifically, the chosen binders are i) Epoxy resin, ii) Sodium Silicate (NaSi) in conjunction with Kaolinite, and iii) Portland cement (refer to table 1). The specimens were labeled as sample R, sample N, and sample C, respectively, based on the binding material employed. Epoxy resin is a two-part synthetic thermosetting polymer known for its excellent adhesion, toughness, and chemical resistance. Cement, on the other hand, is used to bind building blocks like bricks and stones. Lastly, Sodium Silicate is a naturally occurring, water-soluble silicate mineral that can be used as a binding material due to its cementing properties.

2.2. Construction Procedure

To evaluate the efficacy of the selected binding materials, three distinct samples were prepared, each utilizing a different type of binding material. The volume fraction of the binding material in each sample was maintained at 25 %. Detailed information regarding the volume fraction of each binding material can be found in the table 1.

Sodium metasilicate and Kaolinite were mixed in predetermined proportions to form a homogenous slurry. Subsequently, ceramic grains were added to the slurry and carefully hand-mixed with visual inspection to ensure an even distribution of the binding material. Once the components were homogeneously mixed, the resulting slurry was poured into a 3-inch diameter and 8-inch length metal cylinder with movable end caps. The top plate of the cylinder was then subjected to a confining pressure of 1 MPa to compress the ceramic grains. It should be noted that no consideration was given to the loading rate or duration of the confining pressure.

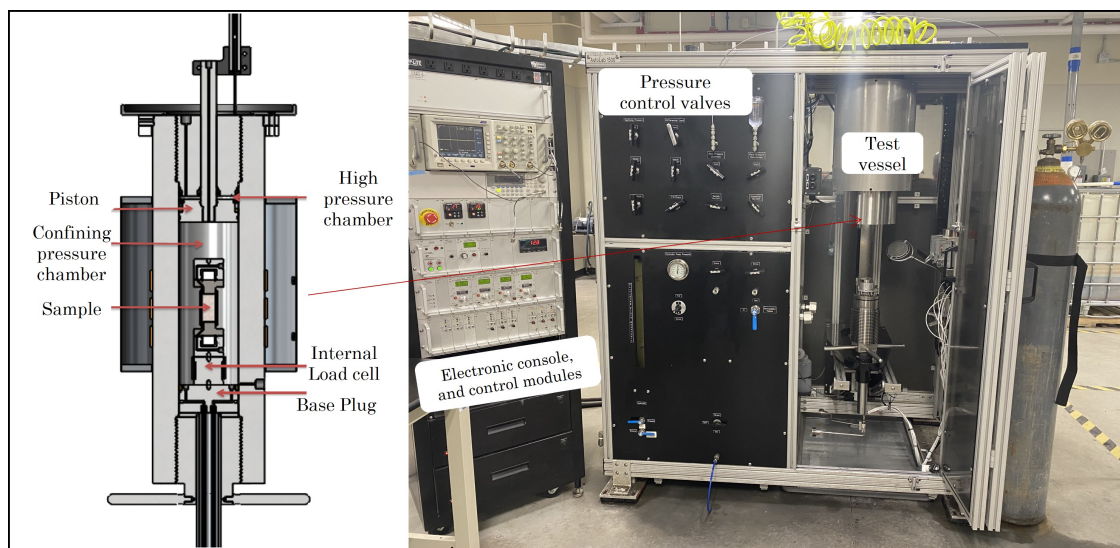


Fig. 1: Figure showing the AutoLab-1500 equipment, located at the University of North Dakota.

Table 1: Binding material volume proportions

Sample	Binding Agent	Volume proportions
Sample-R	Epoxy Resin	Grains (75%) + Epoxy Resin (12.5%) + Epoxy hardner (12.5%)
Sample-N	Soidium Meta Silicate	Grains (75%) + Nasi (15%) + Kaolinite (10%)
Sample-C	Cement	Grains (75%) + Mortar Cement (25%) + Water (7 %)

A similar procedure was followed for the other two binding materials. The binding slurry was prepared first, followed by the addition of ceramic grains, and applying the same confining pressure to ensure a meaningful comparison.

Following the application of confining pressure, the metal cylinders were placed in a furnace, as per the method described by Muñoz-Ibáñez et al. (2020), and heated gradually to a maximum temperature of 300°C at a rate of 50°C per day. Subsequently, the samples were removed from the furnace and allowed to cool down to room temperature. The samples prepared using epoxy resin and cement were not heated, but instead, kept at room temperature for 5 days. After that, all the samples are kept at the room temperature for 30 days. Afterward, the 1" core plugs are extracted from the samples that were synthesized in the metal cylinders using a 1-inch diamond drill bit. Following extraction, all core plugs are subsequently exposed to a pre-heated oven maintained at a temperature of 100°C to eliminate any residual moisture present within the plugs. The final core plugs ready to be tested are shown in figure 2.

2.3. Experiments

We have used the AutoLab-1500 facility to run the acoustic and mechanical experiments, located at the University

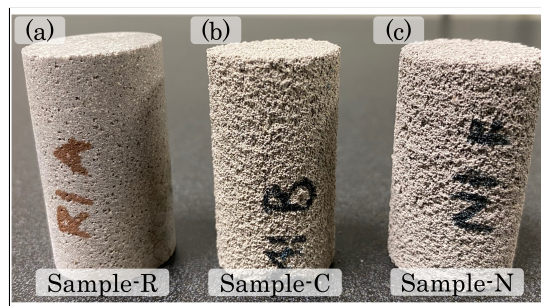


Fig. 2: Three artificial core plugs (a) sample prepared with epoxy resin, (b) sample prepared with cement and (c) sample prepared with Sodium Meta Silicate

of North Dakota, supplied by New England Research, Inc. AutoLab 1500 is a system that uses servo-hydraulics to apply different stress levels on rock samples up to 2 inches wide and tests can be performed under different conditions of pore pressure, and temperature (figure 1). The system enables the execution of conventional rock mechanics testing protocols, such as isotropic compression, simple shear, uniaxial compression, triaxial compression, viscoelastic deformation, elastic waveform measurements, and steady-state and transient permeability measurements. The system has a pressure vessel with a piston inside for changing the stress and hydraulic pumps for adjusting the pressure around and inside the rock. The system can work with

Table 2: Sample dimensions and density

Sample ID	Diameter (mm)	Length (mm)	L/D	Dry bulk density (g/cc)
R	24.02	52.54	2.19	1.84
N	25.45	53.23	2.09	1.62
C	24.79	49.04	1.98	1.66

two different pressure ranges up to 10,000 or 20,000 psi and temperatures up to 248 degF (AutoLab 1500, New England Research, 2017).

2.3.1 Specimen preparation

High-quality physical rock property measurements begin with careful sample preparation. Cylindrical samples of 1-inch diameter and 2-inch length are drilled using a diamond core drill. The exact dimensions of the core plugs extracted are given in table 2. After that, the core plugs are dried in the oven to remove the moisture content and then the sample dry bulk density is measured. Greater care has been taken to ensure that the ends of the specimen are ground flat and parallel. In general, preparing ends of porous sandstone and limestone test specimens for velocity measurements can be difficult. We encountered that the grinding of the surfaces even with the best efforts resulted in the plucking of grains and rough surfaces. Copper foil is used to jacket the test specimen as shown in figure 3. A small confining pressure is applied on the copper jacket by placing the jacketed assembly in the pressure vessel to deform the jacket such that it tightly holds the specimen. The copper jacketed specimen is inserted between the transducer assemblies as shown in figure 3. Small sections of rubber tubing are used for the pedestal of the transducer assembly below the end of the copper jacket.

2.3.2 Ultrasonic wave velocity measurements

The fundamental constituents of an ultrasonic wave velocity measurement apparatus consist of ultrasonic transducers, signal generators, pulse receivers, and data acquisition systems. The specimen under investigation is positioned between the transmitting and receiving transducers, through which ultrasonic waves are propagated (as shown in Figure 3). The wave velocity through the sample material can be determined by evaluating the time of flight between the transmission and reception of ultrasonic signals. The ends of the transducer platens are coated with small amounts of shear wave couplant before placing the sample.

2.3.3 Triaxial test

The ultrasonic wave velocity measurement apparatus is also utilized for triaxial testing. The axial and radial strain gauges employed in this study were provided by HBM Inc. USA. Following the encasement and compression of the copper jacketed core sample, the strain gauges were firmly affixed to the sample via the application of super glue. Subsequently, the strain gauges were connected to the data acquisition system for calibration and further utilization. The triaxial testing protocol was executed following the ISRM guidelines (Feng et al., 2019), which allowed for the measurement of both axial and radial stresses as well as axial and radial deformations. These measured values were subsequently utilized in the computation of various static mechanical properties of the rock sample, such as Young's modulus, bulk modulus, shear modulus, and Poisson's ratio.

2.3.4 Uniaxial compressive strength

The uniaxial compressive strength (UCS) is the utmost axial compressive stress that a right-cylindrical material sample can sustain prior to experiencing failure. UCS is performed on the three samples at a fixed confining pressure of 0.3 MPa. During the UCS test, a constant strain rate of 0.006mm/sec is applied while recording the axial stress-axial strain curve. The test is stopped when a failure is observed in the stress-strain plot and the maximum recorded stress is considered as the UCS of that specimen.

3 RESULTS AND DISCUSSION

The present study aimed to investigate the effect of cementation on the mechanical properties of sandstone. The results obtained from the three experiments, namely ultrasonic wave velocity measurement, triaxial compression test, and unconfined compressive strength (UCS) are presented in the table 3 and plotted in the figures 4, 5 and 6. Figure 2 shows that the distribution of binding material differs among the three samples, indicating a primary limitation in achieving uniform conditions. Additionally, it is also observed that the binding material within each sample is not uniformly distributed. It was observed that Young's modulus and Poisson's ratio were computed at a confining pressure of 4 MPa for all three samples in the triaxial compression experiment (figure 4). Sample R showed the highest Young's modulus and the highest Poisson's ratio among all three samples, whereas sample C displayed the lowest Young's modulus, and sample N showed the lowest Poisson's ratio. This can be attributed to the effect of cementation, as the sediment particles in sandstone become more rigid and less deformable when cemented together by mineral precipitation. The interlocking between

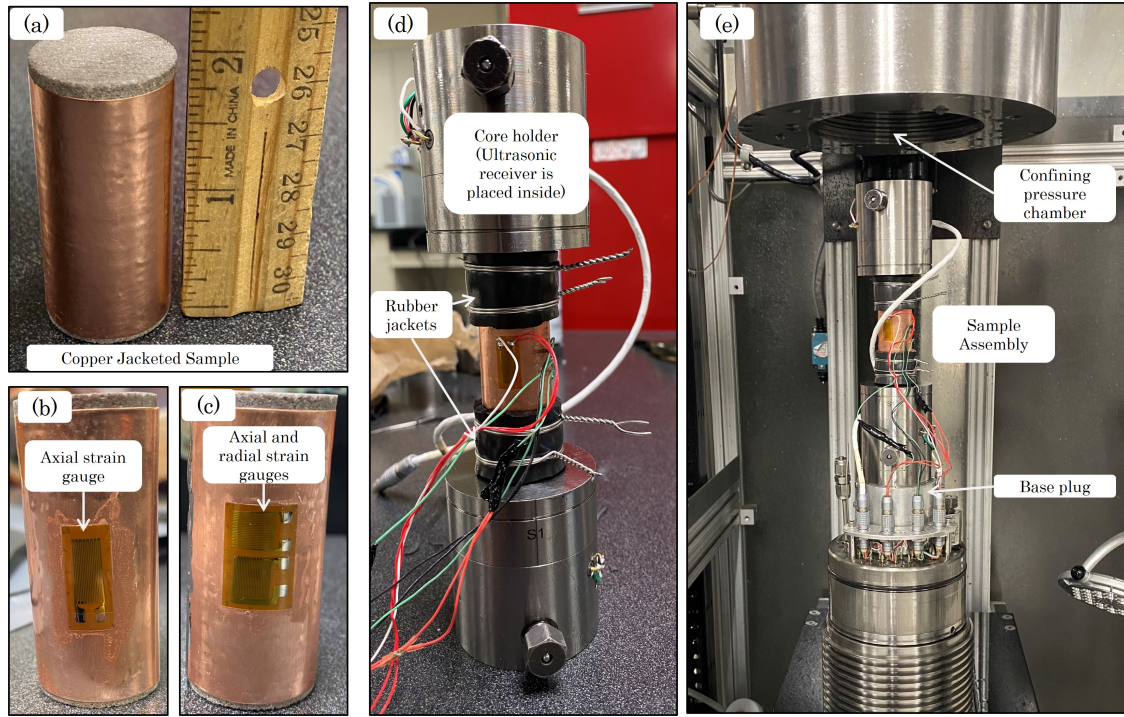


Fig. 3: Sample Assembly (a) copper jacketed sample (b) axial strain gauge (c) radial strain gauge (d) sample is placed in between two core holders (e) sample along with the core holder is placed on the loading frame of the Autolab-1500.

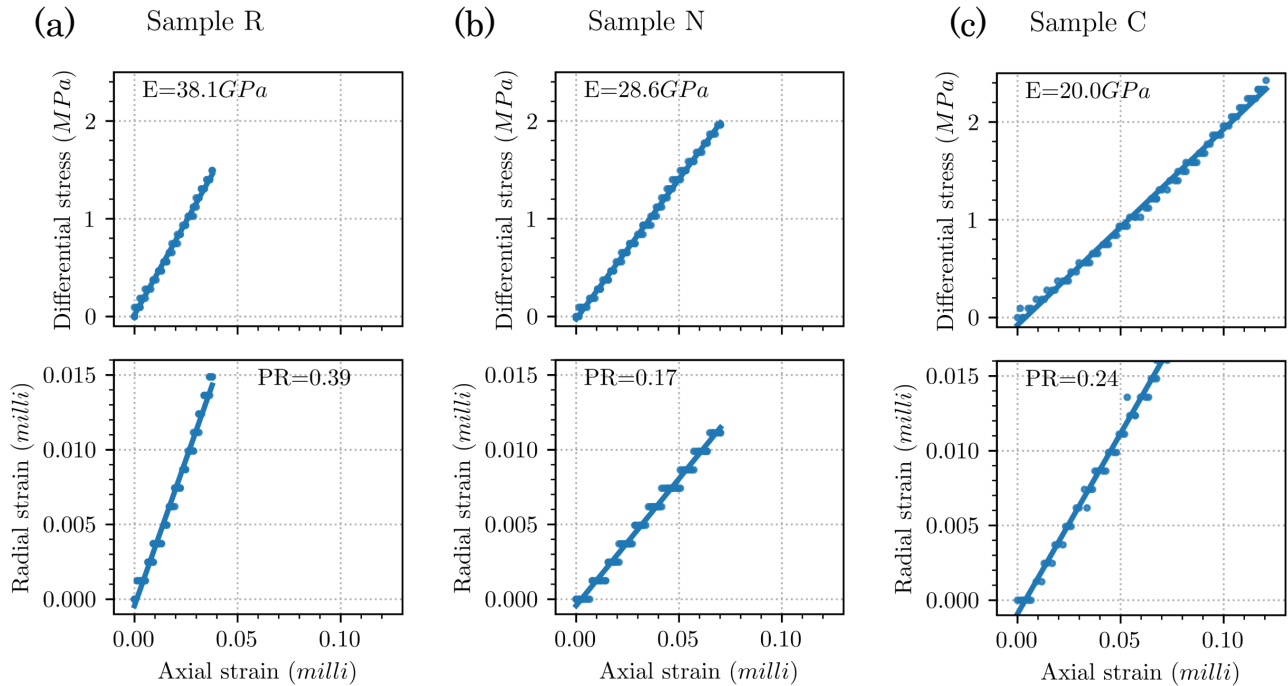


Fig. 4: Traiaxial testing results showing Young's modulus and Poisson's ratio calculations along with the recorded data for the three samples (a) Sample R, (b) Sample N (c) Sample C. It is to be noted that all these measurements are taken at a fixed confining pressure of 4 MPa and at room temperature.

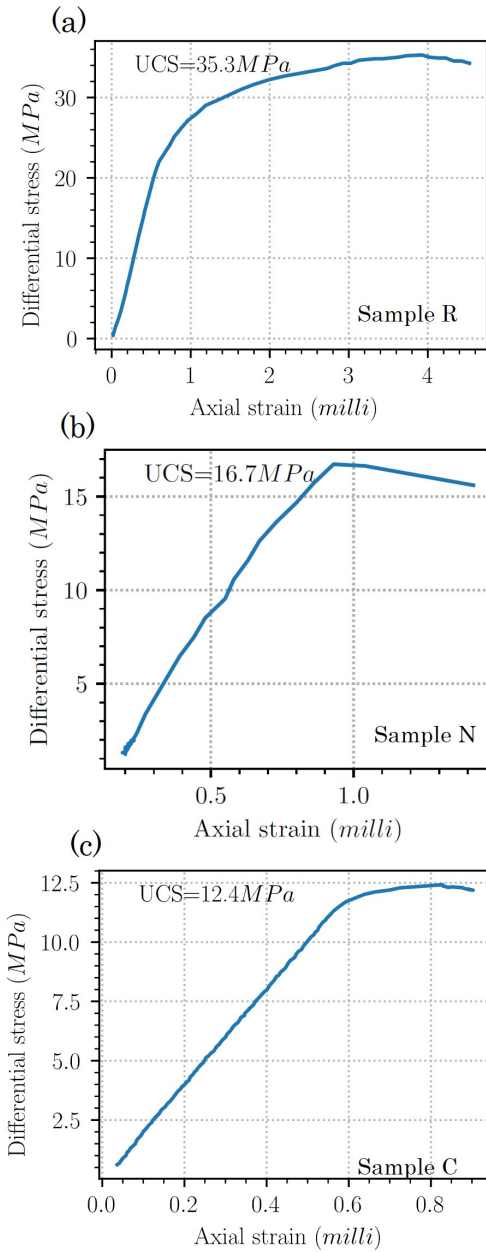


Fig. 5: UCS test results (a) Sample-R, (b) Sample-N (c) Sample-C. UCS is performed at at constant confining pressure of 0.3 MPa

the grains prevents them from moving easily under stress, making the rock stiffer. However, the effect of cementation on Young's modulus can also depend on the type and amount of cementing material present in the rock, as well as the microstructure and porosity of the sandstone.

Further analysis of the results suggested that the epoxy resin-bonded grains appear to be tightly interlocked and display strong resistance to the applied stress compared with the other two samples. Additionally, the relatively higher bulk density of sample R suggests that more binding material is distributed among the grains, reducing

the porosity (figure 2) and increasing the interlocking of grains, thereby increasing Young's modulus.

Similarly, the effect of cementation on Poisson's ratio can also depend on the type and amount of cementing material present in the rock, as well as the microstructure and porosity of the sandstone. The high Poisson's ratio of the epoxy resin probably contributed to the high Poisson's ratio of sample R, as the Poisson's ratio of epoxy resin is strongly dependent on the cure state and generally in between 0.32-0.5 (Saseendran et al., 2017).

The UCS experimental data (figure 5) showed that the UCS of sample R is much higher than that of sample-N and sample-C, indicating that the distribution of resin among the grains and strong interlocking could be the primary reason for the high UCS of sample-R.

It is important to note that the plastic region in the stress-strain deformation curve is not similar among the three samples. The plastic region is important in engineering and materials science, as it determines the ability of a material to withstand deformation and failure under stress. The present results suggest that the epoxy resin binding material display elasto-plastic behaviour and has the largest plastic region, indicating that the material undergoes large deformation before ultimate failure happens. This plastic behaviour is somewhat similar to that of salt rocks (Liang et al., 2012). Sample N shows brittle behaviour and failure. Sample C displays an elasto-plastic behavior with intermediate plastic region among the three samples. However, sample C has the lowest UCS value.

The experimental results demonstrate that the compressional wave velocities of the three rock specimens under investigation exhibit distinct variations, with sample-N exhibiting the highest and sample-C exhibiting the lowest compressional wave velocity (figure 6). P-wave velocity is influenced by various factors, such as the density, porosity, and elasticity of the grains and cement, as well as the quality of the cementation between the grains. It is notable that despite the static Young's modulus and uniaxial compressive strength of sample-N being lower than those of sample-C, the former exhibits a high compressional wave velocity.

4 CONCLUSIONS

- We have prepared three artificial granular rocks with three binding materials (i) Epoxy resin (ii) Sodium Meta Silicate with Kaolinite (iii) Cement and carried out ultrasonic, triaxial, and unconfined confining strength tests.
- The sample prepared with epoxy resin exhibited the

Table 3: Testing results of the three samples

Sample ID	Binding Material	Young's Modulus (GPa)	Poisson's Ratio	Shear Modulus (GPa)	Bulk Modulus (GPa)	UCS (MPa)	$V_p(m/s)$
R	Epoxy Resin	38.10	0.39	13.71	57.73	35.30	3350.0
N	NaSi	28.60	0.17	12.22	14.44	16.70	3698.0
C	Cement	20.10	0.23	8.17	12.41	12.40	2712.0

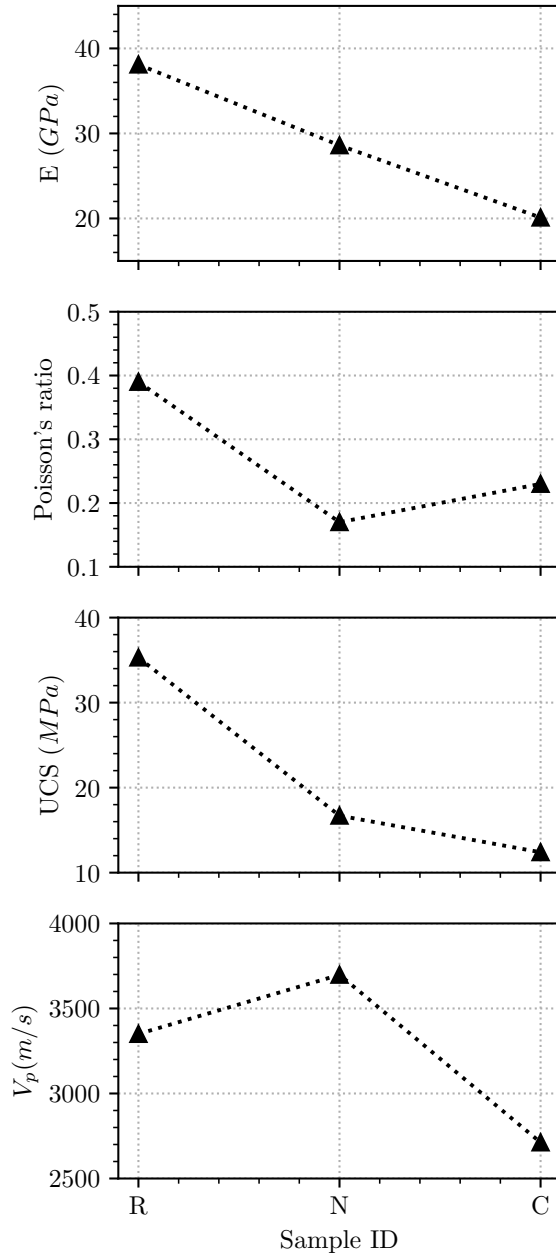


Fig. 6: Comparison of Young's modulus, Poisson's ratio, UCS, and compressional wave velocities among the three samples.

highest values for Young's modulus, Poisson's ratio, and UCS among the three samples.

- The sample prepared with cement showed poor mechanical properties, with the lowest Young's modulus and UCS values among the three samples. The samples prepared with silica gel display moderate behavior in terms of elasticity and strength among the three samples.
- The compressional wave velocity was observed to be highest for the sample prepared with silica gel, whereas the sample prepared with cement exhibited the lowest compressional wave velocity.
- This study emphasizes the influence of cementation on the mechanical properties of artificial sandstones. The results suggest that the type and quantity of cementing material in the rock, as well as the microstructure and porosity of the sandstone, can significantly affect the mechanical behavior of the rock.
- The study's findings indicate that the microstructural properties of the rocks could offer greater insight into their mechanical and strength properties. For instance, techniques such as SEM or micro-CT could reveal the heterogeneous distribution of the binding material within and among the different samples. Further studies are currently underway with the aim of gaining a better understanding of the binding material and micro parameters involved in preparing synthetic rocks that exhibit the desired rock properties.

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